LAB # 6

IMPLEMENTATION OF FOR AND NESTED FOR LOOPS

Lab Tasks

1- Write a program to generate a series of first 50 even numbers.

2- Write a program that generates a table of any number.

3- Write a program to calculate the factorial of any number.

4- what will the output of the following code

```
#include <iostream>
using namespace std;
int main()
{
  for(int i=1; i<=50; i++)
  {
    /* This statement would be executed
     * repeatedly until the condition
     * i<=50 returns false.
     */
     cout<<"Value of variable i is: "<<i-i+2*i*i<<endl;
  }
  return 0;
}
```

5- Write a program that prints following as output.

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
:
:
9 9 9 9 9 9 9 9 9
```

6- Use nested for loop to generate the following as output.

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

7- Write a program that prints a conversion table from miles to kilometers or from kilometers to miles. The program must first ask what kind of conversion table the user wants. After having asked this, you need an **if** construct in the program. the program must use for loop to print the conversion table. the program must print at least 15 conversion lines.

1.Write a program to generate a series of first 50 even numbers.

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    for (int i = 1; i <= 50; i++) {
        if (i % 2 == 0) {
            cout << i<<"\n";
        }
    }
}
```

```
Muhammad Ali
2025F-BIT-023
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

2.Write a program that generates a table of any number.

```cpp
#include <iostream>
using namespace std;
int main()
{
    int num;
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    cout << "Enter number for table: ";
    cin >> num;
    for (int i = 1; i <= 10; i++) {
            cout <<num <<" x "<<i<<" = " << i * num << "\n";
    }
}
```

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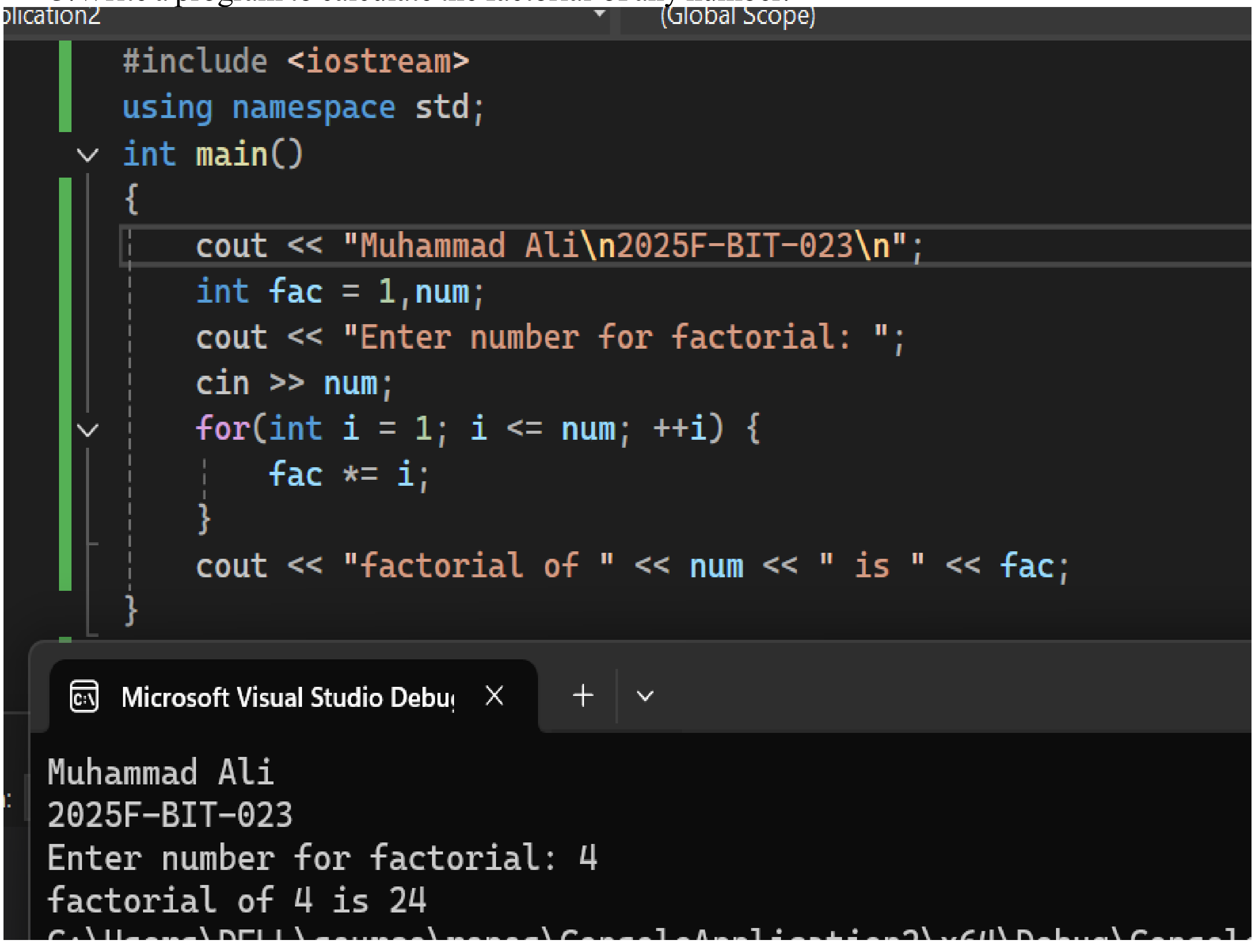
```
Muhammad Ali
2025F-BIT-023
Enter number for table: 2
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```

3.Write a program to calculate the factorial of any number.



4.what will the output of the following code

```
#include <iostream>
        using namespace std;
        int main()
        {
          for(int i=1; i<=50; i++)
          {
            /* This statement would be executed
             * repeatedly until the condition
             * i<=50 returns false.
             */
             cout<<"Value of variable i is: "<<i-i+2*i*i<<endl;
          }
          return 0;
        }
```

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali \n2025F-BIT-023\n";
    for (int i = 1; i <= 50; i++)
    {
        /* This statement would be executed
         *   repeatedly until the condition
         *   i<=50 returns false.
         */
        cout << "Value of variable i is: " << i - i + 2 * i * i << endl;
    }
    return 0;
}
```

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```
Muhammad Ali
2025F-BIT-023
Value of variable i is: 2
Value of variable i is: 8
Value of variable i is: 18
Value of variable i is: 32
Value of variable i is: 50
Value of variable i is: 72
Value of variable i is: 98
Value of variable i is: 128
Value of variable i is: 162
Value of variable i is: 200
Value of variable i is: 242
Value of variable i is: 288
Value of variable i is: 338
Value of variable i is: 392
Value of variable i is: 450
Value of variable i is: 512
Value of variable i is: 578
Value of variable i is: 648
Value of variable i is: 722
Value of variable i is: 800
Value of variable i is: 882
Value of variable i is: 968
Value of variable i is: 1058
Value of variable i is: 1152
Value of variable i is: 1250
Value of variable i is: 1352
Value of variable i is: 1458
Value of variable i is: 1568
Value of variable i is: 1682
Value of variable i is: 1800
Value of variable i is: 1922
Value of variable i is: 2048
Value of variable i is: 2178
Value of variable i is: 2312
Value of variable i is: 2450
Value of variable i is: 2592
Value of variable i is: 2738
Value of variable i is: 2888
Value of variable i is: 3042
Value of variable i is: 3200
Value of variable i is: 3362
Value of variable i is: 3528
Value of variable i is: 3698
Value of variable i is: 3872
Value of variable i is: 4050
Value of variable i is: 4232
Value of variable i is: 4418
Value of variable i is: 4608
Value of variable i is: 4802
Value of variable i is: 5000
```

5.Write a program that prints following as output.

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
:
:
9 9 9 9 9 9 9 9 9

```cpp
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali \n2025F-BIT-023\n";
    for (int i = 1; i <= 9; i++) {
        for (int j = 1; j <= i; j++) {
            cout << i;
        }
        cout <<endl;
    }
}
```

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```
Muhammad Ali
2025F-BIT-023
1
22
333
4444
55555
666666
7777777
88888888
999999999
```

6.Use nested for loop to generate the following as output.

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali \n2025F-BIT-023\n";
    for (int row = 1; row <= 5; row++) {
        for (int col = 1; col <= 5; col++) {
            cout << row*col<<" ";
        }
        cout << endl;
    }
}
```

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```
Muhammad Ali
2025F-BIT-023
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20 25
```

7.Write a program that prints a conversion table from miles to kilometers or from kilometers to miles. The program must first ask what kind of conversion table the user wants. After having asked this, you need an **if** construct in the program. the program must use for loop to print the conversion table. the program must print at least 15 conversion lines.

```
#include <iostream>
using namespace std;
int main()
{
    start:
    cout << "Muhammad Ali \n2025F-BIT-023\n";
    int table;
    cout << "Choose your table:\n1 for Miles to Kilometer \n2 for Kilometer to Miles\n";
    cin >> table;
    if (table == 1) {
        cout << "Miles to Kilometer\n";
        for (int i = 1; i <= 15;i++) {
            cout << i <<" miles is equal to " << i * 1.609 <<" kilometer " << endl;
        }
    }
    else if (table == 2 ) {
        cout << "Kilometer to Miles\n";
        for (int j = 1; j <= 15; j++) {
            cout << j << " kilometer is equal to " << j / 1.609 << " miles " << endl;
        }
    }
    else {
        cout << "invalid option\n";
        goto start;
    }
}
```

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```
Muhammad Ali
2025F-BIT-023
Choose your table:
1 for Miles to Kilometer
2 for Kilometer to Miles
1
Miles to Kilometer
1 miles is equal to 1.609 kilometer
2 miles is equal to 3.218 kilometer
3 miles is equal to 4.827 kilometer
4 miles is equal to 6.436 kilometer
5 miles is equal to 8.045 kilometer
6 miles is equal to 9.654 kilometer
7 miles is equal to 11.263 kilometer
8 miles is equal to 12.872 kilometer
9 miles is equal to 14.481 kilometer
10 miles is equal to 16.09 kilometer
11 miles is equal to 17.699 kilometer
12 miles is equal to 19.308 kilometer
13 miles is equal to 20.917 kilometer
14 miles is equal to 22.526 kilometer
15 miles is equal to 24.135 kilometer
```